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PROTECTIVE DECK COVERING DEVICE, SYSTEM, AND METHOD

Abstract

A composite cover configured for use on a deck board is provided. The composite cover can include a top surface and a bottom surface disposed opposite the top surface. The bottom surface can have a plurality of grooves extending along a distance in a direction of the composite cover. A first side can connect the top surface and the bottom surface. The first side can run longitudinally along a length of the composite cover. A second side can be disposed opposite the first side and connect the top surface and the bottom surface. A composite cover system configured for use on a deck board is also provided. The system can include the composite cover and a fastener. The composite covering system is easy to install, cost-effective, and durable requiring minimal maintenance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/561,104, filed on Mar. 4, 2024. The entire disclosure of the above application is incorporated herein by reference.

FIELD

[0002] The present technology relates generally to devices, systems and methods of construction materials and, more particularly, to a composite cover system.

INTRODUCTION

[0003] This section provides background information related to the present disclosure which is not necessarily prior art.

[0004] Wooden decking materials are used for their natural appearance and ease of installation. However, wooden decking materials and associated surfaces are subject to the whims of nature and wooden decks require significant maintenance due to their susceptibility to environmental damage. Over time, wood can warp, rot, discolor, splinter, and deteriorate, necessitating frequent applications of coatings and stains to preserve its integrity and appearance. This maintenance is not only time-consuming but also expensive and labor-intensive, often requiring reapplication every few years to maintain the deck's weathering properties.

[0005] Composite decking materials emerged as an alternative to wood, offering improved resistance to some of the natural elements that affect wood. Made from thermoplastic polymers and other composites, these materials can require less maintenance and can provide longer life spans. Despite these advantages, composite decks are not without their own issues, such as fading and swelling due to weathering, and they still require maintenance such as sealing, waxing, and the application of protective coverings to enhance aging resistance.

[0006] Deck coverings can improve the life of decking materials and associated surfaces. Certain adhesives can be used to attach such deck coverings, however, adhesives can degrade over time, especially under varying weather conditions, leading to the eventual failure of the deck covering. Moreover, the preparation required for optimal adhesion, such as pressure washing and buffing, adds to the labor and cost of installation.

[0007] Interlocking decking cover systems are another approach to deck coverings, where cover members are made from materials like rigid polyvinyl chloride (PVC). These systems, while providing a degree of protection, can suffer from issues related to the differential thermal expansion between the PVC material and the wooden deck boards. This can lead to cracking and other forms of damage, and the interlocking design may impede proper water drainage, potentially causing water accumulation and mold growth.

[0008] Accordingly, there is a continuing need for a composite cover system that is easy to install, cost-effective, and durable while requiring minimal maintenance.

SUMMARY

[0009] In concordance with the instant disclosure, a composite cover system that is easy to install, cost-effective, and durable while requiring minimal maintenance, has surprisingly been discovered. The present technology includes articles of manufacture, systems, and processes that relate to a composite cover system for use on a deck.

[0010] In certain embodiments, a composite cover configured for use on a deck board is provided. The composite cover can include a top surface and a bottom surface disposed opposite the top surface. The bottom surface can have a number of grooves extending along a distance in a direction of the composite cover. A first side can connect the top surface and the bottom surface. The first side can run longitudinally along a length of the composite cover. A second side can be disposed opposite the first side and connect the top surface and the bottom surface.

[0011] In another embodiment, a composite cover configured for use on a deck board is provided. The composite cover can include a top surface and a bottom surface disposed opposite the top surface. The bottom surface can have a number of grooves extending along a distance in a direction of the composite cover. The composite cover can further include a first end, a second end opposing the first end, a first side extending from the first end, and a second side extending from the second end. The first side and the second side are oriented substantially perpendicular to the top surface. The first side can include a first lip and the second side can include a second lip. The first lip and the second lip can face one another and define a retaining region below the bottom surface.

[0012] In certain embodiments, a fastener for securing a composite cover to a deck board is provided. The fastener can include a fastener body. The fastener body can include a fastener head disposed at a first end and a conical tip disposed at a second end. A first section can be disposed adjacent the conical tip and can be threaded in a first direction. A second section can be disposed between the fastener head and the first section. The second section can be threaded in a second direction.

[0013] In certain embodiments, a composite cover system configured for use on a deck board is provided. The composite cover system can include a composite cover as described herein. The composite cover system can further include a fastener for securing a composite cover to a deck board as described herein.

[0014] In certain embodiments, a method of installing a composite cover system on a board of a deck is provided. The method can include a step of providing a composite cover as described herein. The method can include a step of providing a fastener. The fastener can be a fastener as described herein. The method can include a step of positioning the composite cover on the board of the deck. The method can include a step of securing the composite cover to the board of the deck with the fastener. The method can further include a step of removing a portion of the board to receive a portion of the composite cover. The step of removing the portion of the board can include removing a top surface cover running a longitudinal axis of the board.

[0015] Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

[0016] The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

[0017] FIG. **1** is a perspective schematic illustration of a deck system;

[0018] FIG. **2** is a top perspective view of a composite cover for use with a deck, according to an embodiment of the present disclosure;

[0019] FIG. **3**A is a cross-sectional view of the composite cover, according to the embodiment shown in FIG. **2**;

[0020] FIG. **3**B is an enlarged cross-sectional view of the composite cover, according to the embodiment shown in FIG. **2**;

[0021] FIG. **4** is a bottom perspective view of the composite cover, according to an embodiment shown in FIG. **2**;

[0022] FIG. **5** is a bottom plan view of the composite cover, according to an embodiment shown in FIG. **2**;

[0023] FIG. **6** is a cross-sectional view of a composite cover, according to an embodiment of the present disclosure;
[0024] FIG. **7** is a cross-sectional view of a composite cover, according to an embodiment of the present disclosure;
[0025] FIG. **8** is a bottom perspective view of a composite cover, according to an embodiment of the present disclosure;
[0026] FIG. **9** is a bottom plan view of the composite cover, according to an embodiment shown in FIG. **8**;
[0027] FIG. **10** is a side elevational view of a fastener, according to an embodiment of the present disclosure;
[0028] FIG. **11** is a top plan view of the fastener, according to the embodiment shown in FIG. **10**;
[0029] FIG. **12**A is a top perspective view of a composite cover for use with a deck, according to an embodiment of the present disclosure;
[0030] FIG. **12**B is a bottom perspective view of a composite cover for use with a deck, according to an embodiment of the present disclosure;
[0031] FIG. **13** is a cross-sectional view of the composite cover, according to the embodiment shown in FIG. **12**;
[0032] FIG. **14** is a cross-sectional view of a deck board with a top surface corner removed;
[0033] FIG. **15** is a cross-sectional view of the composite cover of FIG. **12** installed on the deck board of FIG. **14**;
[0034] FIG. **16** is top perspective view of the composite cover of FIG. **12** installed on a portion of the deck board of FIG. **14**;
[0035] FIG. **17** is a top perspective view of a composite cover system, according to an embodiment of the present disclosure;
[0036] FIGS. **18-19** are a flow chart illustrating a method of installing a composite cover system on a board of a deck, according to an embodiment of the present disclosure; and
[0037] FIGS. **20-21** are a flow chart illustrating a method of installing a composite cover system on a concrete surface, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0038] The following description of technology is merely exemplary in nature of the subject matter, manufacture and use of one or more inventions, and is not intended to limit the scope, application, or uses of any specific invention claimed in this application or in such other applications as may be filed claiming priority to this application, or patents issuing therefrom. Regarding methods disclosed, the order of the steps presented is exemplary in nature, and thus, the order of the steps can be different in various embodiments, including where certain steps can be simultaneously performed, unless expressly stated otherwise. “A” and “an” as used herein indicate “at least one” of the item is present; a plurality of such items may be present, when possible. Except where otherwise expressly indicated, all numerical quantities in this description are to be understood as modified by the word “about” and all geometric and spatial descriptors are to be understood as modified by the word “substantially” in describing the broadest scope of the technology. “About” when applied to numerical values indicates that the calculation or the measurement allows some slight imprecision in the value (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If, for some reason, the imprecision provided by “about” and/or “substantially” is not otherwise understood in the art with this ordinary meaning, then “about” and/or “substantially” as used herein indicates at least variations that may arise from ordinary methods of measuring or using such parameters.
[0039] Although the open-ended term “comprising,” as a synonym of non-restrictive terms such as including, containing, or having, is used herein to describe and claim embodiments of the present technology, embodiments may alternatively be described using more limiting terms such as “consisting of” or “consisting essentially of.” Thus, for any given embodiment reciting materials,

components, or process steps, the present technology also specifically includes embodiments consisting of, or consisting essentially of, such materials, components, or process steps excluding additional materials, components or processes (for consisting of) and excluding additional materials, components or processes affecting the significant properties of the embodiment (for consisting essentially of), even though such additional materials, components or processes are not explicitly recited in this application. For example, recitation of a composition or process reciting elements A, B and C specifically envisions embodiments consisting of, and consisting essentially of, A, B and C, excluding an element D that may be recited in the art, even though element D is not explicitly described as being excluded herein.

[0040] As referred to herein, all compositional percentages are by weight of the total composition, unless otherwise specified. Disclosures of ranges are, unless specified otherwise, inclusive of endpoints and include all distinct values and further divided ranges within the entire range. Thus, for example, a range of "from A to B" or "from about A to about B" is inclusive of A and of B. Disclosure of values and ranges of values for specific parameters (such as amounts, weight percentages, etc.) are not exclusive of other values and ranges of values useful herein. It is envisioned that two or more specific exemplified values for a given parameter may define endpoints for a range of values that may be claimed for the parameter. For example, if Parameter X is exemplified herein to have value A and also exemplified to have value Z, it is envisioned that Parameter X may have a range of values from about A to about Z. Similarly, it is envisioned that disclosure of two or more ranges of values for a parameter (whether such ranges are nested, overlapping or distinct) subsume all possible combination of ranges for the value that might be claimed using endpoints of the disclosed ranges. For example, if Parameter X is exemplified herein to have values in the range of 1-10, or 2-9, or 3-8, it is also envisioned that Parameter X may have other ranges of values including 1-9, 1-8, 1-3, 1-2, 2-10, 2-8, 2-3, 3-10, 3-9, and so on.

[0041] When an element or layer is referred to as being "on," "engaged to," "connected to," or "coupled to" another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to" or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0042] Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0043] Spatially relative terms, such as "inner," "outer," "beneath," "below," "lower," "above," "upper," and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" the other elements or features. Thus, the example term "below" can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0044] In accordance with the illustrated embodiments of the present disclosure, a composite cover **100**, a fastener **200**, a composite cover system **300**, another composite cover **400**, and a method **500** of installing the composite cover system **300** are provided. Advantageously, the present disclosure addresses shortcomings in outdoor living spaces by providing a composite cover **100** and composite cover system **300** for use on a deck **10** that enhances durability and incorporates effective liquid and vapor management features to militate liquid and vapor accumulation and associated problems. The present disclosure minimizes the need for maintenance by providing a composite cover **100** that resists fading, splintering, and warping, eliminating the need for pressure washing, sanding, staining, or sealing, while shielding against UV rays, moisture, and harsh weather.

[0045] Referring now to the drawings, illustrated in FIG. **1** is a diagrammatic and simplified view of aspects used in construction of a deck **10**. The deck **10** can be constructed in various ways, as known to one skilled in the art. The deck **10** can include a plurality of support-posts **12** (or more simply "posts" or individually "a post"), a plurality of joists **14** (individually "a joist"), a bridging **16**, and a plurality of deck boards **18** (individually "a deck board" or "board"). The plurality of joists **14** extend between the plurality of support-posts **12**, a bridging **16** is positioned between the joists **14** for added stability, and a plurality of deck boards **18** supported by the plurality of joists **14** form a deck surface **20**. It should be appreciated that the deck **10** can vary in design, material, and construction without departing from the scope of the present disclosure. For example, the dimensions, spacing, and arrangement of the support-posts **12**, joists **14**, bridging **16**, and deck boards **18** can be modified to accommodate different load requirements, environmental conditions, or aesthetic preferences such as overall design or shape. Additionally, while the deck **10** is illustrated in a simplified form, it may include additional structural or decorative elements, such as railings, stairs, or fasteners, as used in the art.

[0046] In certain embodiments, and with reference to FIGS. **2-5**, a composite cover **100** for use on a deck board **18** can include a top surface **102** and a bottom surface **104** disposed opposite the top surface **102**. The bottom surface **104** can include one or more grooves **106** formed thereon. The grooves **106** can extend along a distance **108** in a direction **110** of the composite cover **100**. The grooves **106** will be discussed in more detail below. The composite cover **100** can include a first side **112**. The first side **112** can connect the top surface **102** and the bottom surface **104** and can run longitudinally along a length **114** of the composite cover **100**. The composite cover **100** can include a second side **116** disposed opposite the first side **112**. The second side **116** can connect the top surface **102** to the bottom surface **104** and can run longitudinally along the length **114** of the composite cover **100**.

[0047] With reference to FIGS. **2-4**, the composite cover **100** can include a transition **118***a* formed at a junction **120***a* of the top surface **102** and the first side **112**. The composite cover **100** can include another transition **118***b* formed at a junction **120***b* of the top surface **102** and the second side **116**. Similarly, the composite cover **100** can include a transition **118***c* formed at a junction **120***c* of the bottom surface **104** and the first side **112**, and another transition **118***d* formed at a junction **120***d* of the bottom surface **104** and the second side **116**. The transitions **118***a*, **118***b*, **118***c*, **118***d* can include a smooth, curved profile, such as a fillet or a radius, or can include a chamfer or a beveled edge. The transitions **118***a*, **118***b*, **118***c*, **118***d* can help shed liquid more efficiently, reducing liquid and moisture retention, and minimizing the risk of rotting or warping over time. Additionally, the transitions **118***a*, **118***b*, **118***c*, **118***d* can be less prone to chipping or splitting compared to sharp corners, as the curved profile can distribute impact forces more evenly, enhancing the composite cover's durability. Furthermore, the transitions **118***a*, **118***b*, **118***c*, **118***d* can accommodate slight expansion and contraction due to temperature and humidity changes, helping to maintain the composite cover's **100** structural integrity while militating against excessive wear.

[0048] The top surface **102** and the bottom surface **104** can include texturing **122**. The texturing **122** can provide several advantages, including increased friction, masking scratches, dents, and signs of wear while helping to hide dirt and stains, reducing maintenance efforts. Additionally, texturing **122** can minimize glare by diffusing light. There are several types of texturing **122**, each serving different functional and aesthetic purposes. A wood grain texture can mimic the natural patterns and ridges of real wood, offering a traditional, authentic appearance. Wood grain textures can contribute to durability by reducing the visibility of surface wear while maintaining a natural and inviting look without the maintenance demands of traditional wood. Brushed finishes feature fine streaks for a subtle, modern look, while grooved or ribbed surfaces incorporate linear patterns to enhance traction and improve liquid drainage. The texturing **122** will be discussed in more detail below.

[0049] With reference to FIGS. **2-9**, incorporating one or more grooves **106** on the bottom surface **104** of the composite cover can help manage liquid and vapor, thereby enhancing the longevity of the underlying deck **10**. The grooves **106** can include a cross-section designed for liquid and vapor management, directing rainwater, spilled liquids, or moisture condensation away from a surface of the deck. The grooves **106** can help militate against water pooling, which can lead to mold or mildew growth, while also promoting airflow to facilitate drying and reduce warping and/or decay of the deck. Additionally, the grooves **106** can allow vapor to escape, minimizing the moisture being trapped between the composite cover **100** and the deck surface **20**. The grooves **106** can each have various cross-sectional shapes, including a U-shape, a V-shape, a trapezoidal shape, a right trapezium shape, or combinations thereof.

[0050] The shape and spacing of the grooves **106** can also contribute to weight reduction of the composite cover **100**. By incorporating grooves **106** into the design of the composite cover **100**, less material can be used when manufacturing, which directly reduces the overall weight of the composite cover **100**. This lighter weight can help make the composite cover **100** easier to handle, transport, and install, especially when placing it on top of the deck **10**. While reducing the material volume, the grooves **106** can be designed to maintain the composite cover's **100** structural integrity. The placement and shape of the grooves **106** can help ensure that the composite cover **100** remains strong and capable of supporting loads without warping, cracking, or bending. For example, using grooves **106** can remove material in a way that preserves the composite cover's **100** strength, allowing it to withstand the stresses it will encounter during use. The reduced weight can be achieved without compromising the composite cover's **100** ability to provide the necessary performance characteristics, such as load-bearing capacity and resistance to wear, ensuring that the composite cover **100** performs reliably over time.

[0051] The grooves **106** can each have a width **124** and a height **126**. The width **124** can be between 1 millimeter and 50 millimeters. The height **126** can be between 1 millimeter and 7 millimeters. The width **124** and height **126** of the grooves **106** can be chosen such that the remaining solid areas of the composite cover **100** can continue to bear load, ensuring that the composite cover **100** is capable of supporting weight and resisting deformation under stress. Furthermore, the grooves **106** can be uniformly spaced apart between 1 millimeter and 50 millimeters to ensure even liquid distribution and optimized airflow. This spacing of the grooves **106** not only helps with functional aspects like drainage and ventilation but also contributes to maintaining the structural integrity of the composite cover **100**. The distribution of grooves **106** can help ensure that the forces exerted on the composite cover **100**, whether from foot traffic, furniture, or environmental pressures, are spread out, maintaining the composite cover's **100** overall strength, rigidity, and durability.

[0052] With reference to FIGS. **4-5** and **8-9**, the grooves **106** can extend along a distance **108** in a direction **110** of the composite cover **100**. The direction **110** can align with either a longitudinal axis **128** or a lateral axis **130** of the composite cover **100**. The orientation of the grooves **106** can be chosen to optimize liquid drainage and airflow Additionally, the grooves **106** can extend along the distance **108** across the composite cover **100**, which can encompass the entire length **114** of the composite cover **100** or only a portion of it.

[0053] The composite cover **100** can be characterized by a length **132**, a width **134**, and a height **136**. The length **132** of the composite cover **100** can vary to accommodate different decking or structural applications, with options ranging from 6, 8, 12, 16 to 20 feet. The width **134** of the composite cover **100** can range from 4 to 7 inches. A narrower width **134** may be suited for more intricate or compact designs, while a wider width **134** allows for broader coverage, reducing the number of boards required for a particular surface area. The height **136** can range from 5 to 20 millimeters. This variation in height **136** accommodates different thicknesses for the composite cover **100**, providing options for strength, durability, and performance. Thicker composite covers **100** may be more suitable for high-traffic areas or environments with extreme weather conditions, while thinner composite covers **100** offer lighter weight and ease of installation without sacrificing structural integrity. By offering a range of dimensions for length **132**, width **134**, and height **136**, the composite cover **100** can meet both functional and aesthetic requirements. It should be understood that one having ordinary skill in the art can select a suitable length, width, and height of the composite cover **100** within the scope of the present disclosure.
[0054] In certain embodiments, and with refence to FIGS. **10** and **11**, a fastener **200** for securing a composite cover **100** to a deck board **18** is provided. The fastener **200** can include a fastener body **202**. The fastener body **202** can include a fastener head **204** disposed at a first end **206** and a conical tip **208** disposed at a second end **210**. A first section **212** can be disposed adjacent the conical tip **208** and can be threaded in a first direction **214**. A second section **216** can be disposed between the fastener head **204** and the first section **212** and can be threaded in a second direction **218**. The second direction **218** can be different from the first direction **214**. The fastener **200** can reduce the risk of cracking or deformation in the composite cover **100**, improving the fastener's **200** ability to securely anchor in various substrates with minimal effort.
[0055] The fastener head **204** can be equipped with a drive cavity **220** for secure engagement with a driving tool. The drive cavity **220** can be a T15 Torx drive, which can optimize torque tolerance and minimize the risk of stripping during installation. The drive cavity's **220** dimensions can include a diameter of 5.5±0.1 mm and a depth of 2.8±0.1 mm, which can help driving tool engagement and minimize slippage. To further enhance the fastener's **200** durability and color matching with the composite cover **100**, the fastener head **204** can be treated with a baked enamel color coating, which can provide corrosion resistance. It should be understood that one having ordinary skill in the art can select a suitable type of drive cavity **220** within the scope of the present disclosure. Examples can include Hex, Torx, TR Torx, Phillips and Slotted Bit.
[0056] The first section **212** can have a first length **222** that can be greater than a second length **224** of the second section **216**. The first section **212** can include a self-tapping triangular split thread, which can allow the fastener **200** to cut into the composite cover **100** and deck **10** as it is driven in. This self-tapping feature can enable the fastener **200** to engage in the composite cover **100** securely without the need for pre-drilling. The second section **216**, positioned between the fastener head **204** and the first section **212**, can include a self-tapping reverse thread, which can provide additional stability during installation and facilitate easier driving into harder materials. The second section **216** of the fastener **200** can include a self-tapping reverse-threaded ST4.5 thread, which can transition to a self-tapping ST4.0 triangular split thread in the first section **212**. The fastener **200** can reduce driving resistance, balance grip force, and minimize installation torque.
[0057] When securing the composite cover **100** to an underlying board **18**, the fastener **200** can be positioned at a predetermined position with the conical tip **208** on the composite cover **100**. A driving tool, such as a drill or screwdriver, can be used to rotate the fastener **200**, advancing it through the composite cover **100**, and into the underlying board **18**. As the fastener **200** is driven in, the first section having the self-taping triangular split threads continues from the composite cover **100** into the underlying board **18**. As the fastener **200** is further driven in, the second section **216** having the self-tapping reverse-threads can engage with the composite cover **100**. The reverse threads can create a secure grip within the composite cover **100**, anchoring the fastener **200** and

militating against any tendency for the fastener **200** to loosen or back out over time.

[0058] The combination of self-taping triangular split threads and the self-tapping reverse-threads can increase the holding strength of the fastener **200**. The reverse threads can prevent the fastener **200** from backing out by counteracting forces that might otherwise loosen it due to environmental factors or structural movement. As a result, the composite cover **100** can remain securely affixed to the board **18**, ensuring long-term stability and durability.

[0059] In certain embodiments, and with reference to FIG. **17**, a composite cover system **300** for use on a deck **10** is provided. The composite cover system **300** can include a composite cover **100** and a fastener **200**. The composite cover **100** can be the composite cover **100** as described hereinabove. The fastener can be any fastener, or the fastener **200** as described hereinabove.

[0060] In certain embodiments, with reference to FIGS. **12-13** and **15-16**, a composite cover **400** for use on a deck board **18** can include a top surface **402** and a bottom surface **404** disposed opposite the top surface **402**. The bottom surface **404** can have a groove **406** formed thereon. The groove **406** can extend along a distance **408** in a direction **410** of the composite cover **400**.

[0061] The composite cover **400** can include a first end **412**, a second end **414** opposing the first end **412**, a first side **416** extending from the first end **412**, and a second side **418** extending from the second end **414**. The first side **416** and the second side **418** can be oriented substantially perpendicular to the top surface **402** and the first side **416** can include a first lip **420** and the second side **418** can include a second lip **422**. The first lip **420** and the second lip can each include inside edges **423***a*, **432***b*. The inside edges **423***a*, **432***b* can face one another and define a retaining region **424** below the bottom surface **404**.

[0062] With reference to FIG. **13**, the composite cover **400** can include a transition **426***a* formed at a junction **428***a* of the top surface **402** and the first side **416**. The composite cover **400** can include another transition **426***b* formed at a junction **428***b* of the top surface **402** and the second side **418**. The transitions **426***a*, **426***b* can have a smooth, curved profile, such as a fillet or a radius, or include a chamfer or a beveled edge. The transitions **426***a*, **426***b* can help shed liquid more efficiently, reducing liquid and moisture retention, and minimizing the risk of rotting or warping over time. Additionally, the transitions **426***a*, **426***b* can be less prone to chipping or splitting compared to sharp corners, as the curved profile can distribute impact forces more evenly, enhancing the composite cover's **400** durability. Furthermore, the transitions **426***a*, **426***b* can accommodate slight expansion and contraction due to temperature and humidity changes, helping to maintain the composite cover's **400** structural integrity while militating against excessive wear.

[0063] With reference to FIG. **12**A, the top surface **402** can include texturing **430**. The texturing **430** can provide several advantages, including increased friction, masking scratches, dents, and signs of wear while helping to hide dirt and stains, reducing maintenance efforts. Additionally, the texturing **430** can minimize glare by diffusing light. There are several types of texturing, each serving different functional and aesthetic purposes. A wood grain texture can mimic the natural patterns and ridges of real wood, offering a traditional, authentic appearance. Wood grain textures can contribute to durability by reducing the visibility of surface wear while maintaining a natural and inviting look without the maintenance demands of traditional wood. Brushed finishes feature fine streaks for a subtle, modern look, while grooved or ribbed surfaces incorporate linear patterns to enhance traction and improve liquid drainage. The texturing **430** will be discussed in more detail below.

[0064] With reference to FIGS. **12-13** and **15-16**, incorporating one or more grooves **406** on the bottom surface **404** of the composite cover **400** can help manage liquid and vapor, thereby enhancing the longevity of the underlying deck **10**. The grooves **406** can include a cross-section designed for liquid and vapor management, directing rainwater, spilled liquids, or moisture condensation away from the deck surface **20**. The grooves **406** can help militate against water pooling, which can lead to mold or mildew growth, while also promoting airflow to facilitate drying and reduce warping and/or decay of the deck **10**. Additionally, the grooves **406** can allow vapor to escape, minimizing the moisture being trapped between the composite cover **400** and the deck surface **20**. The grooves **406** can each have various cross-sectional shapes, including but not limited to, a U-shape, a V-shape, a trapezoidal shape, a right trapezium shape, or combinations thereof.

[0065] The specific shape and spacing of the grooves **406** can also contribute to weight reduction of the composite cover **400**. By incorporating grooves **406** into the design of the composite cover **400**, less material can be used when manufacturing, which directly reduces the overall weight of the composite cover **400**. This lighter weight can help make the composite cover **400** easier to handle, transport, and install, especially when placing it on top of the deck **10**. While reducing the material volume, the groove configuration is designed to maintain the composite cover's **400** structural integrity. The placement and shape of the grooves **406** can help ensure that the composite cover **400** remains strong and capable of supporting loads without warping, cracking, or bending. For example, using grooves **406** can remove material in a way that preserves the composite cover's **400** strength, allowing it to withstand the stresses it will encounter during use. The reduced weight can be achieved without compromising the composite cover's **400** ability to provide the necessary performance characteristics, such as load-bearing capacity and resistance to wear, ensuring that the composite cover **400** performs reliably over time.

[0066] The grooves **406** can be similar to the grooves **106** shown in FIGS. **2-9**. The grooves **406** can have a width **124** and a height **126**. The width **124** can be between 1 millimeter and 50 millimeters. The height **126** can be between 1 millimeter and 7 millimeters. The width **124** and the height **126** of the grooves **406** can be chosen such that the remaining solid areas of the composite cover **400** can continue to bear load, ensuring that the composite cover **400** is capable of supporting weight and resisting deformation under stress. Furthermore, the grooves **406** can be uniformly spaced apart between 1 millimeter and 50 millimeters to ensure even liquid distribution and optimized airflow. This spacing of the grooves **406** not only helps with functional aspects like drainage and ventilation but also contributes to maintaining the structural integrity of the composite cover **400**. The distribution of grooves **406** can help ensure that the forces exerted on the composite cover **400**, whether from foot traffic, furniture, or environmental pressures, are spread out, maintaining the composite cover's **400** overall strength, rigidity, and durability.

[0067] With reference to FIGS. **12**A & **12**B, the grooves **406**, can extend along a distance **408** in a direction **410** of the composite cover **400**. The direction **140** can align with either a longitudinal axis **436** or a lateral axis **438** of the composite cover **400**, depending on the desired design and functional requirements. The orientation of the grooves **406** can be chosen to optimize liquid drainage, airflow, and aesthetic appearance. Additionally, the grooves **406** can extend along the distance **408** across the composite cover **400**, which can encompass the entire length of the composite cover or only a portion of it.

[0068] The composite cover **400** can have a length **440**, a width **442**, and a height **444**. The length **440** of the composite cover **400** can vary to accommodate different decking or structural applications, with options ranging from 6, 8, 12, 16 to 20 feet. The width **442** of the composite cover **400** can range from 4 to 7 inches. A narrower width may be suited for more intricate or compact designs, while a wider width allows for broader coverage, reducing the number of composite covers required for a particular surface area. The height **444** can range from 5 to 20 millimeters. This variation in height **444** can accommodate different thicknesses for the composite cover **400**, providing options for strength, durability, and performance. Thicker composite covers **400** may be more suitable for high-traffic areas or environments with extreme weather conditions, while thinner options offer lighter weight and ease of installation without sacrificing structural integrity. By offering a range of dimensions for length **440**, width **442**, and height **444**, the composite cover **400** can be meet both functional and aesthetic requirements. It should be understood that one having ordinary skill in the art can select a suitable length, width, and height of the composite cover **400** within the scope of the present disclosure.

[0069] The composite cover **100**, **400** can be manufactured using an extrusion process, which can provide a consistent and efficient method for producing high-quality, dimensionally stable composite covers **100**, **400**. In the extrusion process, a mixture of thermoplastic polymers, reinforcing fibers, and additives can be heated and blended into a homogeneous material. The heated composite material can then be forced through a die that shapes it into the desired profile, ensuring uniform thickness and surface characteristics.

[0070] During extrusion, additional features can be incorporated on the composite cover **100**, **400**, such as surface texturing (e.g., wood grain patterns) or co-extruded protective layers, which can enhance the composite cover's **100**, **400** resistance to UV exposure, moisture, and surface wear. The extrusion process can also allow for precise control over the material composition, ensuring that the final product meets structural and aesthetic requirements. Furthermore, by using continuous extrusion, long sections of composite covers can be produced efficiently, reducing material waste and improving production scalability. Texturing **122**, **430** can be imprinted onto a composite cover **100**, **400** using various manufacturing techniques that enhance both its aesthetic appeal and functional performance. One method can include embossing during the extrusion process, where a patterned roller or plate applies pressure to the surface of the heated composite material as it exits the die. This technique can create a consistent texture, such as a wood grain pattern, grooves, or ridges, giving the composite cover **100**, **400** a natural look while improving slip resistance. The depth and sharpness of the texture can be controlled by adjusting the pressure and temperature during embossing, ensuring a durable and long-lasting surface.

[0071] Another approach involves co-extrusion, where an additional protective layer is applied to a composite core during manufacturing. This additional protective layer can include an integrated texture, such as a deep-grain wood pattern or a roughened anti-slip surface. Co-extrusion can provide additional scratch resistance, UV protection, and enhanced color retention, making it ideal for outdoor applications where the board is exposed to weathering and foot traffic. The co-extruded layer can also help reduce maintenance by preventing surface fading, staining, and moisture absorption.

[0072] For other textures, press molding or stamping can be used. In these methods, pre-formed molds with the desired texture are pressed onto the composite cover's **100**, **400** surface before it fully cools. This technique allows for intricate detailing, making it useful for replicating the complex grain patterns of natural wood or achieving unique, decorative designs.

[0073] The composite cover **100**, **400** can be formed from a composite material formulated for durability, weather resistance, and structural integrity. The composite material can include a thermoplastic polymer, to provide flexibility and moisture resistance. It can also include a wood material to enhance strength and reduce thermal expansion. To improve material performance, the composite can incorporate a coupling agent, which can enhance the adhesion between the polymer and wood fibers, increasing overall stability.

[0074] The composite material can include various additives, such as an ultra-violet (UV) stabilizer to reduce degradation from sun exposure, an antioxidant to prevent thermal oxidation, and a processing aid to improve manufacturability. A colorant or a pigment can be added for aesthetic customization, while a filler additive can enhance density and mechanical properties. A blowing agent can be included to create a lightweight, foamed structure, reducing material weight while maintaining strength. Furthermore, the composite material can incorporate an impact modifier to improve resistance to cracking and a scratch-resistant additive to enhance surface durability, reducing wear from foot traffic and environmental factors.

[0075] The thermoplastic polymer can include polypropylene, polyethylene, polyvinyl chloride, styrene-butadiene-styrene (SBS), styrene-ethylene-butylene-styrene (SEBS), thermoplastic polyurethanes (TPUs), natural rubber (NR), nitrile rubber (NBR), chloroprene rubber (CR), carboxyl-terminated butadiene acrylonitrile (CTBN), recycled polypropylene, recycled polyethylene, recycled polyvinyl chloride, rubber granule, recycled rubber granule, and

combinations thereof.
[0076] The wood material can include sawdust, wood flour, finely milled hardwood, finely milled softwood, wood fiber, and lignocellulosic fibers, wood chips, microcrystalline cellulose, nanocellulose, torrefied wood particles, bark powder, lignin powder, and combinations thereof.
[0077] The coupling agent can include maleic anhydride-grafted polyethylene/polypropylene (MAPP), maleic anhydride grafted styrene-ethylene-butylene-styrene, silane based such as aminopropyltriethoxysilane and vinyltriethoxysilane, diisocyanate based such as polymeric methylene diphenyl diisocyanate (pMDI), functionalized polyolefins such as epoxidized polypropylene, and combinations thereof.
[0078] The ultra-violet (UV) stabilizer can include hindered amine light stabilizers (HALS), such as Bis(2,2,6,6-tetramethyl-4-piperidyl) sebacate (BTMPS) (Tinuvin® 770) and Poly [[6-[(1,1,3,3-tetramethylbutyl)amino]-1,3,5-triazine-2,4-diyl][(2,2,6,6-tetramethyl-4-piperidinyl)imino]-1,6-hexanediyl [(2,2,6,6-tetramethyl-4-piperidinyl)imino]]) (Chimassorb® 944), 2,4-Dihydroxybenzophenone (Uvinul® 400), Bisoctrizole (Tinosorb® M), and combinations thereof.
[0079] The process aid can include polyethylene wax, stearates, stearic acid, metal stearates such as calcium stearate and zinc stearate, amide waxes such as ethylene Bis(Stearamide) (EBS), paraffin wax, and FT-wax, hydrocarbon waxes such as low-molecular-weight polyethylene, polypropylene wax, recycled waxes, ester-based process aids, fatty acid amides such as erucamide and oleamide, and combinations thereof. The process aid can also include a plasticizer including phthalates such as diisononyl phthalate (DINP) and diisodecyl phthalate (DIDP), non-phthalates such as dioctyl adipate (DOA), epoxidized soybean oil, acetyl tributyl citrate (ATBC), and chlorinated paraffin, and combinations thereof.
[0080] The colorant/pigment can include titanium dioxide, iron oxide, carbon black, chromium oxide, ultramarine blue, cobalt blue, cadmium pigment, phthalocyanine blue, phthalocyanine green, azo pigment, quinacridone pigment, carbonized wood, biochar, hematoxylin, tannin-based pigment, aluminum powder, pearlescent mica pigment, and combinations thereof.
[0081] The filler additive can include calcium carbonate, talc, kaolin clay, silica, alumina trihydrate (ATH), magnesium hydroxide, barium sulfate, wollastonite, mica, feldspar, perlite, vermiculite, glass beads, glass fibers, ceramic microspheres, fly ash, basalt fibers, cellulose fibers, lignin, starch, biochar, nanocellulose, carbon black, graphite, montmorillonite, sepiolite, dolomite, zinc oxide, titanium dioxide, iron oxide, and combinations thereof.
[0082] The blowing agent can include azodicarbonamide (ADC), sodium bicarbonate, citric acid, hydrazine derivatives, benzene sulfonyl hydrazide, p-toluene sulfonyl hydrazide, p-toluene sulfonyl semicarbazide, 5-phenyltetrazole, urea-based blowing agents, expandable microspheres, isocyanates, hydrocarbons (butane, pentane, isopentane), carbon dioxide, nitrogen, water, hydrofluorocarbons (HFCs), hydrochlorofluorocarbons (HCFCs), hydrofluoroolefins (HFOs), formic acid, and combinations thereof.
[0083] The impact modifier can include ethylene vinyl acetate (EVA), acrylic impact modifiers such as methyl methacrylate butadiene styrenic (MBS) resins, chlorinated polyethylene (CPE) resins, acrylic resins, acrylonitrile butadiene styrene (ABS), styrene-ethylene-butadiene-styrene (SEBS), styrene-butadiene rubber (SBR), ethylene propylene diene monomer (EPDM), ethylene-methyl acrylate (EMA), ethylene-ethyl acrylate (EEA), maleic anhydride-grafted polymers (e.g., MA-g-PE, MA-g-PP), polyolefin elastomers (POE), thermoplastic polyurethanes (TPU), core-shell rubber particles, ionomers (e.g., Surlyn), polysiloxane-based impact modifiers, and combinations thereof.
[0084] Example formulations of the composite cover **100**, **400** are provided in Tables 1-7, shown below, so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. The formulations for the composite cover **100**, **400** can be used in various manufacturing processes, such as extrusion and co-extrusion.
TABLE-US-00001 TABLE 1 Formulation 1 Component % By Weight Thermoplastic Polymer 20-

50% Wood Material 40-80% Coupling Agent 1-6% Ultra-violet (UV) Stabilizer 0.5-2% Antioxidant 0.1-1.5% Processing Aid 1-3% Colorant/Pigment 1-3% Filler Additive 0-5%
TABLE-US-00002 TABLE 2 Formulation 2 Component % By Weight Thermoplastic Polymer 15-30% Wood Material 40-80% Coupling Agent 1-6% Ultra-violet (UV) Stabilizer 0.5-2% Antioxidant 0.1-1.5% Processing Aid 1-3% Colorant/Pigment 1-3% Filler Additive 0-5% Impact Modifier 5-20%
TABLE-US-00003 TABLE 3 Formulation 3 Component % By Weight Thermoplastic Polymer 20-50% Wood Material 40-80% Coupling Agent 1-6% Ultra-violet (UV) Stabilizer 0.5-2% Antioxidant 0.1-1.5% Processing Aid 1-3% Colorant/Pigment 1-3% Filler Additive 0-5% Blowing Agent 0.1-6%
TABLE-US-00004 TABLE 4 Formulation 4 Component % By Weight Thermoplastic Polymer A 50-70% Thermoplastic Polymer B 0-10% Ultra-violet (UV) Stabilizer 1-3% Antioxidant 0.1-0.5% Processing Aid 0.5-2% Colorant/Pigment 2-20% Scratch Resistive Additive 1-2% Impact Modifier 1-4%
TABLE-US-00005 TABLE 5 Formulation 5 Component % By Weight Thermoplastic Polymer 50-60% Ultra-violet (UV) Stabilizer 1-3% Antioxidant 0.1-0.5% Processing Aid 0.5-2% Colorant/Pigment 10-20% Scratch Resistive Additive 1-2% Impact Modifier 2-5% Filler Additive 5-20%
TABLE-US-00006 TABLE 6 Formulation 6 Component % By Weight Thermoplastic Polymer 50-60% Ultra-violet (UV) Stabilizer 1-3% Antioxidant 0.1-0.5% Processing Aid 0.5-2% Colorant/Pigment 10-20% Scratch Resistive Additive 1-2% Impact Modifier 10-30%
TABLE-US-00007 TABLE 7 Formulation 7 Component % By Weight Thermoplastic Polymer A 20-50% Thermoplastic Polymer B 40-80% Processing Aid 1-3% Filler Additive 0-5%
[0085] With reference to FIGS. **17** and **18**, a method **500** of installing a composite cover system **300** on a board **18** of a deck **10** is provided. The method **500** can include a step **502** of providing the composite cover **100**, **400** as described hereinabove. The method can include a step **504** of providing a fastener. The fastener can be any fastener or the fastener as described hereinabove. The method can include a step **506** of preparing the deck **10**. The method can include a step **508** of positioning the composite cover **100**, **400** on the board **18** of the deck **10**. The method **500** can include a step **510** of securing the composite cover **100**, **400** to the board **18** of the deck **10** with the fastener **200**, whereby the composite cover **100**, **400** is installed on the board **18**.
[0086] In certain embodiments, the method **500** can further include a step **512** of removing a portion **446** of the board **18** to receive a portion **448** of the composite cover **400**. The removing of the portion **446** of the board **18** can include removing a top surface corner **450** running along a longitudinal axis of the board **18**. The portion **448** of the composite cover **400** can substantially conform to the removed portion **446** of the board **18**. The method **500** can further include a step **514** of positioning the composite cover **400** on the board **18** of the deck **10**. The method **500** can further include a step **516** of securing the composite cover **400** to the board **18** of the deck **10** with the fastener **200**, whereby the composite cover **400** is installed on the board **18**.
[0087] With reference to FIGS. **19-20**, in certain embodiments, a method **600** of installing a composite cover system **300** on a concrete surface is provided. The method **600** can include a step **602** of providing the composite cover **100** as described hereinabove. The method can include a step **604** of providing a fastener. The fastener can be any fastener or the fastener as described hereinabove. The method **600** can include a step **606** of providing an adhesive. The method **600** can include a step **608** of roughening a bottom surface **104** of the composite cover **100**. The method **600** can include a step **610** of preparing the concrete surface. The method **600** can include a step **612** of positioning the composite cover **100** on the concrete surface. The method **600** can include a step **614** of securing the composite cover **100** to the concrete surface with the adhesive and the fastener **200**.
[0088] Example embodiments are provided so that this disclosure will be thorough, and will fully

convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms, and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail. Equivalent changes, modifications and variations of some embodiments, materials, compositions and methods can be made within the scope of the present technology, with substantially similar results.

Claims

1. A composite cover configured for use on a deck board, comprising: a top surface; a bottom surface disposed opposite the top surface, the bottom surface having a plurality of grooves formed thereon, the plurality of grooves extending along a distance in a direction of the composite cover; a first side connecting the top surface and the bottom surface running longitudinally along a length of the composite cover; and a second side disposed opposite the first side and connecting the top surface and the bottom surface running longitudinally along the length of the composite cover.
2. The composite cover of claim 1, wherein the top surface includes texturing.
3. The composite cover of claim 1, wherein each groove of the plurality of grooves includes a substantially U-shaped cross-section.
4. The composite cover of claim 1, wherein each groove of the plurality of grooves has a width between 1 millimeter and 50 millimeters.
5. The composite cover of claim 1, wherein each groove of the plurality of grooves has a height between 1 millimeter and 7 millimeters.
6. The composite cover of claim 1, wherein each groove of the plurality of grooves are uniformly spaced apart from one another.
7. The composite cover of claim 1, wherein the distance includes an entirety of the length of the composite cover.
8. The composite cover of claim 1, wherein the direction is along a longitudinal axis of the composite cover.
9. The composite cover of claim 1, wherein the composite cover is formed of a composite material comprising: between 20% and 40% of a thermoplastic polymer; between 40% and 80% of a wood material; between 1% and 6% of a coupling agent; between 0.5% and 2% of an ultra-violet stabilizer; between 0.1% and 1.5% of an antioxidant; between 1% and 3% of a process aid; between 1% and 3% of a pigment; and between 0% and 5% of a filler additive.
10. The composite cover of claim 1, wherein the composite cover comprises one or more of the following materials: high-density polyethylene; wood flour; maleic anhydride grafted polyethylene; bis(2,2,6,6-tetramethyl-4-piperidinyl) sebacate; pentaerythritol tetrakis(3-(3,5-di-tert-butyl-4-hydroxyphenyl) propionate); and paraffin wax.
11. A fastener for securing a composite cover to a deck board, comprising: a fastener body, including: a fastener head disposed at a first end; a conical tip disposed at a second end; a first section disposed adjacent the conical tip, the first section threaded in a first direction; and a second section disposed between the fastener head and the first section, the second section threaded in a second direction, different than the first direction.
12. The fastener of claim 11, wherein the first section includes a first length and the second section includes a second length, the first length is greater than the second length.
13. A composite cover system configured for use on a deck board, comprising: a composite cover including: a top surface; a bottom surface disposed opposite the top surface, the bottom surface having a plurality of grooves formed thereon, the plurality of grooves extending along a

longitudinal direction of the composite cover; a first side connecting the top surface and the bottom surface running longitudinally along a length of the composite cover; and a second side disposed opposite the first side and connecting the top surface and the bottom surface running longitudinally along the length of the composite cover; and the fastener of claim 11.

14. A composite cover configured for use on a deck board, comprising: a top surface; a bottom surface disposed opposite the top surface, the bottom surface having a plurality of grooves formed thereon, the plurality of grooves extending along a distance in a direction of the composite cover; a first end; a second end opposing the first end; a first side extending from the first end; and a second side extending from the second end, wherein: the first side and the second side are oriented substantially perpendicular to the top surface and the first side includes a first lip and the second side includes a second lip, the first lip and the second lip each have an inside edge facing one another and define a retaining region below the bottom surface.

15. The composite cover of claim 14, wherein the composite cover is formed of a composite material comprising: between 20% and 40% of a thermoplastic polymer; between 40% and 80% of a wood material; between 1% and 6% of a coupling agent; between 0.5% and 2% of an ultra-violet stabilizer; between 0.1% and 1.5% of an antioxidant; between 1% and 3% of a process aid; between 1% and 3% of a pigment; and between 0% and 5% of a filler additive.

16. The composite cover of claim 14, wherein the top surface includes texturing.

17. A method of installing a composite cover system on a board of a deck, comprising: providing the composite cover of claim 1; providing a fastener, the fastener including: a fastener body, including: a fastener head disposed at a first end; a conical tip disposed at a second end; a first section disposed adjacent the conical tip, the first section threaded in a first direction; and a second section disposed between the fastener head and the first section, the second section threaded in a second direction, different than the first direction; positioning the composite cover on the board of the deck; and securing the composite cover to the board of the deck with the fastener, whereby the composite cover is installed on the board.

18. The method of claim 17, wherein prior to the positioning of the composite cover on the board of the deck, the method further comprises: removing a portion of the board to receive a portion of the composite cover; positioning the composite cover on the board of the deck; and securing the composite cover to the board of the deck with the fastener, whereby the composite cover is installed on the board.

19. The method of claim 18, wherein removing the portion of the board includes removing a top surface corner running along a longitudinal axis of the board.

20. The method of claim 19, wherein the portion of the composite cover substantially conforms to the removed portion of the board.